1 Mask

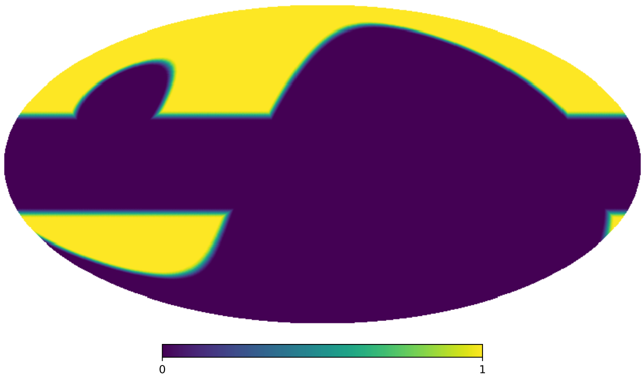

Figure 1: Analysis mask combining the QUIJOTE low-declination survey mask with a 20° cut in galactic latitude.

2 Data

Our analysis employs two complementary datasets: simulated sky maps generated using the `Python` Sky Model (`PySM`) library, which incorporates all major Galactic foreground components; and observational data from the QUIJOTE, WMAP, and Planck experiments. All simulated maps are convolved with the appropriate beam window functions for each respective experiment to ensure realistic instrumental effects.

Before fitting the observed power spectra, we subtract the CMB contribution using the best-fit theoretical ΛCDM spectrum from Planck 2018 (Planck Collaboration, 2020). This spectrum is obtained from the full Plik TTTEEE likelihood combined with low-multipole (lowE) and gravitational lensing constraints. For polarization modes, we subtract both C_ℓ^{EE} and C_ℓ^{BB}, where the latter includes only the contribution from B-modes generated by gravitational lensing, since primordial B-modes of inflationary origin have not yet been detected. This CMB subtraction is essential to isolate the Galactic foreground emission (synchrotron and thermal dust) that constitutes the target of this analysis.

3 Error estimation

Power spectrum uncertainties are estimated through Monte Carlo simulations for both datasets. We compute the standard deviation from 100 realizations each of sky+noise and pure noise maps. The noise characteristics vary by experiment: QUIJOTE and Planck employ the official noise simulations provided by their respective collaborations, incorporating both white noise and correlated $1/f$ noise components. For WMAP, we model the noise as purely white, following standard practice for this dataset.

4 MCMC fitting

We model the total angular power spectrum between two frequency bands ν_i and ν_j as the sum of synchrotron, dust, and their cross-correlation,

$$C_\ell(\nu_i,\nu_j) = C_\ell^{\rm sync}(\nu_i,\nu_j) + C_\ell^{\rm dust}(\nu_i,\nu_j) + C_\ell^{\rm sd}(\nu_i,\nu_j). \tag{1}$$

The synchrotron spectrum is modeled using a power-law frequency scaling and an ℓ-dependent slope,

$$C_\ell^{\rm sync}(\nu_i,\nu_j) = A_s \left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_s} \times \frac{S_s(\nu_i)}{{\rm CC}_s(i,\beta_{s,{\rm cc}})} \frac{S_s(\nu_j)}{{\rm CC}_s(j,\beta_{s,{\rm cc}})}. \tag{2}$$

where $S_s(\nu) = (\nu/\nu_{\rm ref})^{\beta_s}$ and α_s is the multipole (angular) slope. The dust spectrum is described by a modified blackbody (MBB) in Rayleigh–Jeans units with an ℓ-dependent slope,

$$C_\ell^{\rm dust}(\nu_i,\nu_j) = A_d \left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_d} \times \frac{S_d(\nu_i)}{{\rm CC}_d(i,\beta_{d,{\rm cc}})} \frac{S_d(\nu_j)}{{\rm CC}_d(j,\beta_{d,{\rm cc}})}. \tag{3}$$

with

$$S_d(\nu) = \frac{B_\nu(\nu,T_d)}{B_\nu(\nu_0,T_d)} \left(\frac{\nu}{\nu_0}\right)^{\beta_d} \times \frac{g_{\rm RJ}(\nu_0)}{g_{\rm RJ}(\nu)}. \tag{4}$$

where B_ν is the Planck function and $g_{\rm RJ}(\nu) = 2k\nu^2/c^2$ converts to Rayleigh–Jeans temperature. The synchrotron–dust cross term is

$$C_\ell^{\rm sd}(\nu_i,\nu_j) = \rho\sqrt{A_s A_d} \left(\frac{\ell}{\ell_{\rm ref}}\right)^{(\alpha_s+\alpha_d)/2} \times \left[\frac{S_s(\nu_i)}{{\rm CC}_s(i,\beta_{s,{\rm cc}})} \frac{S_d(\nu_j)}{{\rm CC}_d(j,\beta_{d,{\rm cc}})} + \frac{S_s(\nu_j)}{{\rm CC}_s(j,\beta_{s,{\rm cc}})} \frac{S_d(\nu_i)}{{\rm CC}_d(i,\beta_{d,{\rm cc}})}\right]. \tag{5}$$

Color corrections are applied multiplicatively per band and per component via the factors $\mathrm{CC}_s(i, \beta_s)$ and $\mathrm{CC}_d(i, \beta_d)$. These depend on the source frequency spectral index β. Following *fastcc* (Peel et al., 2022), each color correction is represented as a quadratic polynomial in the Rayleigh–Jeans domain variable

$$\alpha_{\mathrm{cc}} = \beta + 2\,, \tag{6}$$

namely

$$\mathrm{CC}(b, \alpha_{\mathrm{cc}}) = a_0 + a_1\,\alpha_{\mathrm{cc}} + a_2\,\alpha_{\mathrm{cc}}^2\,, \tag{7}$$

with band-dependent coefficients $\{a_0, a_1, a_2\}$ for both synchrotron and dust derived from *fastcc*. For the Planck HFI dust bands, polynomials tabulated versus β are evaluated in the α_{cc} domain using the identity above to ensure consistency in Rayleigh–Jeans units.

In the above parameterization, A_s and A_d are the component amplitudes at the reference multipole $\ell_{\mathrm{ref}} = 80$ and reference frequencies $\nu_{\mathrm{ref}} = 11.1\,\mathrm{GHz}$ (synchrotron) and $\nu_0 = 353.0\,\mathrm{GHz}$ (dust), respectively; α_s and α_d are the multipole (angular) spectral indices; β_s and β_d are the frequency spectral indices; T_d is the dust temperature, fixed at $19.6\,\mathrm{K}$; and ρ is the synchrotron–dust correlation coefficient.

4.1 Data Selection and Frequency Coverage

We perform a Markov Chain Monte Carlo (MCMC) fitting procedure using data from QUIJOTE 11 GHz; WMAP 23, 33, 41, 61, and 94 GHz; and Planck 30, 44, 70, 100, 143, 217, and 353 GHz, covering the multipole range $30 < \ell < 200$. In this study, we use both auto- and cross-spectra between frequency bands. This yields 91 independent spectra from the 13 frequency bands.

4.2 Bayesian Inference

We adopt a Bayesian framework to constrain the model parameters $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$. The posterior distribution is given by

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{8}$$

where $\mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})$ is the likelihood and $p(\boldsymbol{\theta})$ is the prior. The likelihood is constructed from the observed power spectra $\hat{C}_\ell(\nu_i, \nu_j)$ and their uncertainties:

$$\ln \mathcal{L} = -\frac{1}{2} \sum_{i,j,\ell} \frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \tag{9}$$

For the power-law model fitting, we adopt uniform (flat) priors for all model parameters within physically motivated bounds:

$$\begin{aligned} A_s &\in [0, \infty), & (10)\\ \alpha_s &\in [-6, 0], & (11)\\ \beta_s &\in [-6, 0], & (12)\\ A_d &\in [0, \infty), & (13)\\ \alpha_d &\in [-6, 0], & (14)\\ \beta_d &\in [0, 6], & (15)\\ \rho &\in [-1, 1], & (16) \end{aligned}$$

For the bin-to-bin analysis, to regularize the fit and incorporate external information from the literature, we impose a Gaussian prior on the synchrotron frequency spectral index:

$$\beta_s \sim \mathcal{N}(-3.1, 0.30^2), \tag{17}$$

while maintaining uniform priors for the remaining parameters within the bounds specified above.

4.3 Fitting methods

We apply three complementary approaches to the angular power spectra computed from the maps:

Power-law fitting, $\Delta\ell = 10$ We compute the spectra with a bin width $\Delta\ell = 10$ and fit a power-law model for the multipole dependence of the primary foreground components (synchrotron and dust). This fit estimates the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), the multipole spectral indices (α_s, α_d), and the correlation coefficient (ρ) between components.

Power-law fitting, $\Delta\ell = 20$ We repeat the power-law fit using a wider binning, $\Delta\ell = 20$, to test the stability of the fitted parameters with respect to the choice of bin size. Wider bins reduce per-bin noise at the cost of multipole resolution.

Bin-to-bin analysis, $\Delta\ell = 20$ In this approach each multipole bin is fitted independently with $\Delta\ell = 20$, without assuming a global power-law dependence on ℓ. For each bin we only fit the amplitudes (A_s, A_d), the frequency spectral indices (β_s, β_d), and the correlation (ρ); the multipole spectral indices (α_s, α_d) are not constrained in this analysis. This method allows us to identify scale-dependent behaviour and local deviations from the power-law assumption.

5 Results

In the following subsections we first present the results obtained on simulated data and then the results for observational data, applying the three procedures described previously (Section 4.3).

5.1 Observational data

We show results for the same set of fitting approaches used on the simulations to allow direct comparison.

5.1.1 Power-law fit ($\Delta\ell = 20$)

The power-law fit with wider bins ($\Delta\ell = 20$) is shown in Figure 2 (EE) and Figure 3 (BB).

5.1.2 Bin-to-bin analysis ($\Delta\ell = 20$)

The bin-to-bin analysis for observational data fits each multipole bin independently; results are listed in Table **??** and illustrated in Figure 4.

References

Peel, M. W., Genova-Santos, R., Dickinson, C., Leahy, J. P., López-Caraballo, C., Fernández-Torreiro, M., Rubiño-Martín, J. A., and Spencer, L. D. (2022). Fastcc: Fast color corrections for broadband radio telescope data. *Research Notes of the AAS*, 6(12):252.

Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *A&A*, 641:A6.

Table 1: Bin-to-bin fit results. EE mode (top) and BB mode (bottom).

ℓ range	$\ell_{\rm eff}$	$A^{\rm EE}_{\rm sync}$ [μK^2]	$\beta^{\rm EE}_{\rm sync}$	$A^{\rm EE}_{\rm dust}$ [10^{-3} μK^2]	$\beta^{\rm EE}_{\rm dust}$	$\rho^{\rm EE}$
20–39	29.0	24.44 ± 0.74	-3.11 ± 0.02	18.10 ± 0.20	1.40 ± 0.01	0.003 ± 0.010
40–59	49.0	3.16 ± 0.23	-3.12 ± 0.06	3.62 ± 0.05	1.43 ± 0.02	0.074 ± 0.015
60–79	69.0	1.15 ± 0.20	-3.20 ± 0.43	1.42 ± 0.13	1.38 ± 0.12	-0.094 ± 0.045
80–99	89.0	0.55 ± 0.12	-3.18 ± 0.45	1.03 ± 0.08	1.41 ± 0.11	0.052 ± 0.025
100–119	109.0	0.16 ± 0.06	-3.01 ± 0.48	0.72 ± 0.03	1.49 ± 0.15	0.069 ± 0.101
120–139	129.0	0.10 ± 0.05	-2.87 ± 0.26	0.38 ± 0.01	1.25 ± 0.04	0.184 ± 0.084
140–159	149.0	0.13 ± 0.05	-2.81 ± 0.30	0.28 ± 0.01	1.37 ± 0.06	0.093 ± 0.059
160–179	169.0	0.15 ± 0.07	-3.16 ± 0.29	0.23 ± 0.01	1.47 ± 0.05	0.049 ± 0.082
180–199	189.0	0.05 ± 0.05	-3.09 ± 0.30	0.17 ± 0.01	1.55 ± 0.07	0.131 ± 0.209
ℓ range	$\ell_{\rm eff}$	$A^{\rm BB}_{\rm sync}$ [μK^2]	$\beta^{\rm BB}_{\rm sync}$	$A^{\rm BB}_{\rm dust}$ [10^{-3} μK^2]	$\beta^{\rm BB}_{\rm dust}$	$\rho^{\rm BB}$
20–39	29.0	3.35 ± 0.34	-3.04 ± 0.07	7.29 ± 0.10	1.44 ± 0.02	0.183 ± 0.021
40–59	49.0	0.80 ± 0.20	-3.15 ± 0.49	2.57 ± 0.03	1.41 ± 0.02	-0.003 ± 0.067
60–79	69.0	0.10 ± 0.06	-3.09 ± 0.27	1.18 ± 0.02	1.53 ± 0.02	0.141 ± 0.109
80–99	89.0	0.04 ± 0.04	-3.11 ± 0.52	0.67 ± 0.03	1.50 ± 0.16	-0.007 ± 0.172
100–119	109.0	0.05 ± 0.03	-2.92 ± 0.43	0.40 ± 0.01	1.45 ± 0.12	0.057 ± 0.149
120–139	129.0	0.06 ± 0.04	-3.09 ± 0.27	0.24 ± 0.01	1.52 ± 0.04	0.121 ± 0.154
140–159	149.0	0.02 ± 0.03	-3.11 ± 0.30	0.21 ± 0.00	1.48 ± 0.04	0.052 ± 0.245
160–179	169.0	0.17 ± 0.08	-3.25 ± 0.26	0.15 ± 0.00	1.44 ± 0.05	-0.085 ± 0.069
180–199	189.0	0.13 ± 0.05	-2.78 ± 0.24	0.12 ± 0.00	1.45 ± 0.06	-0.073 ± 0.070

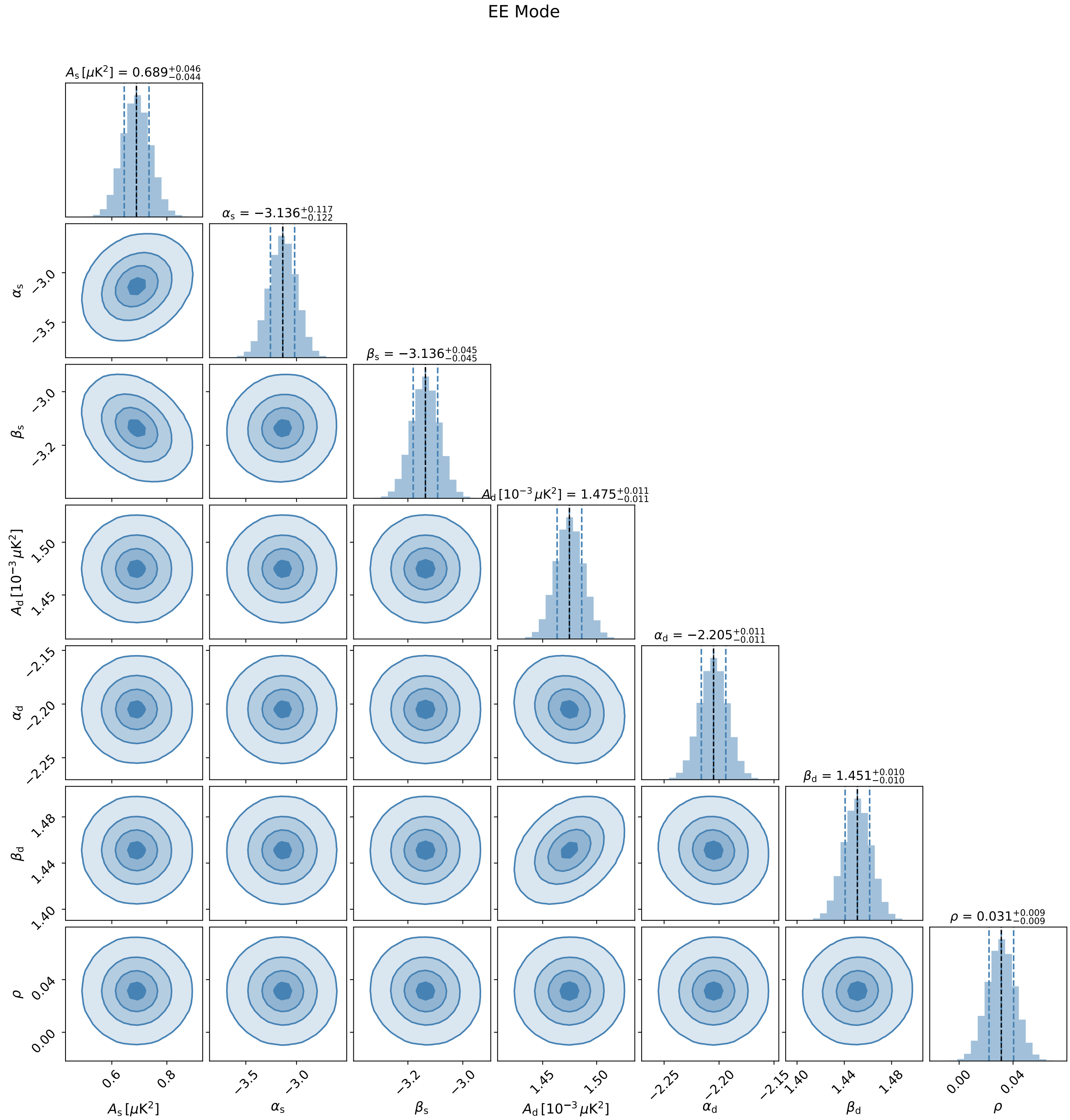

Figure 2: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational EE mode data with bin width $\Delta\ell = 20$.

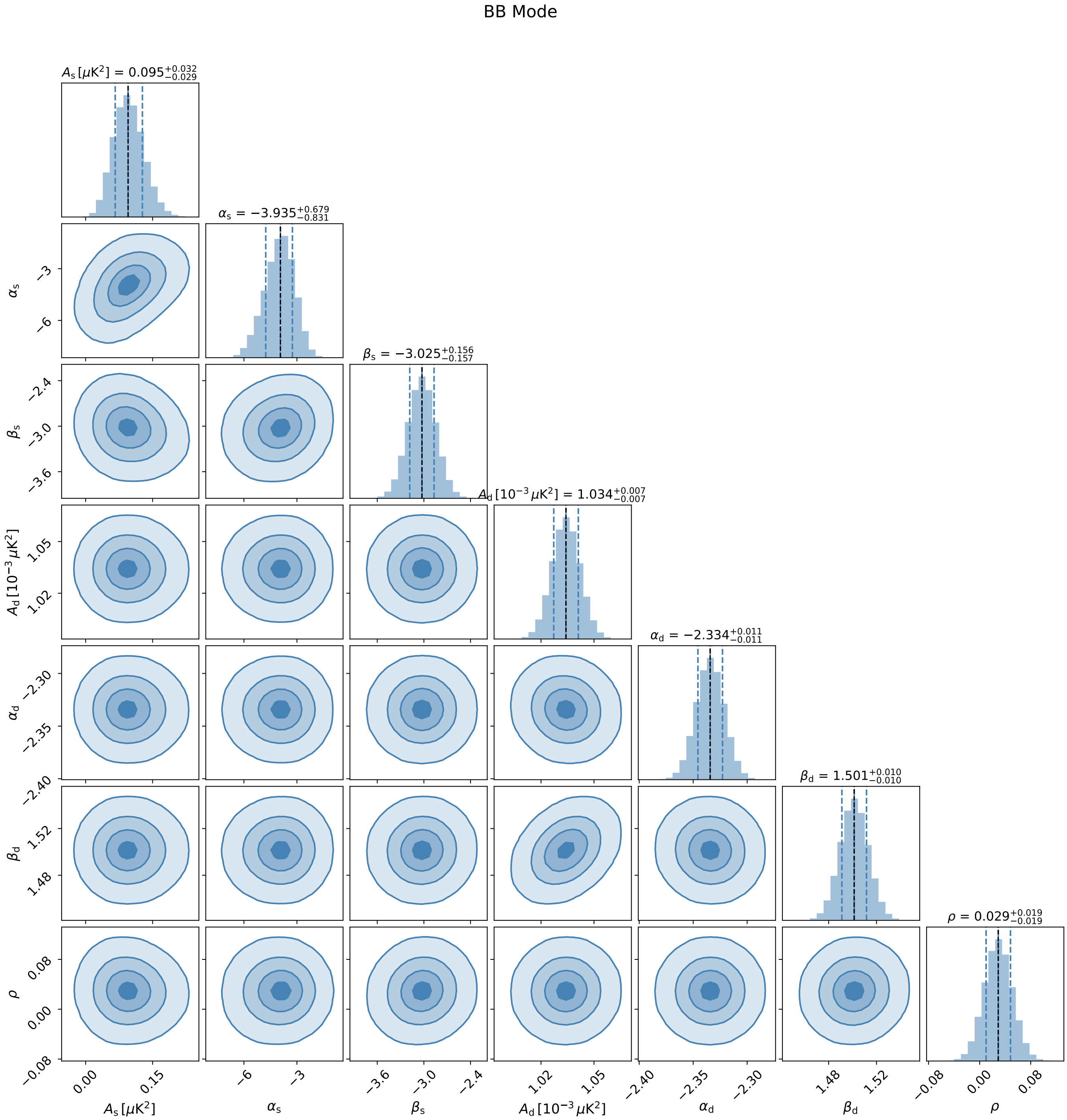

Figure 3: Corner plot showing posterior distributions for synchrotron and dust model parameters from MCMC fitting of observational BB mode data with bin width $\Delta\ell = 20$.

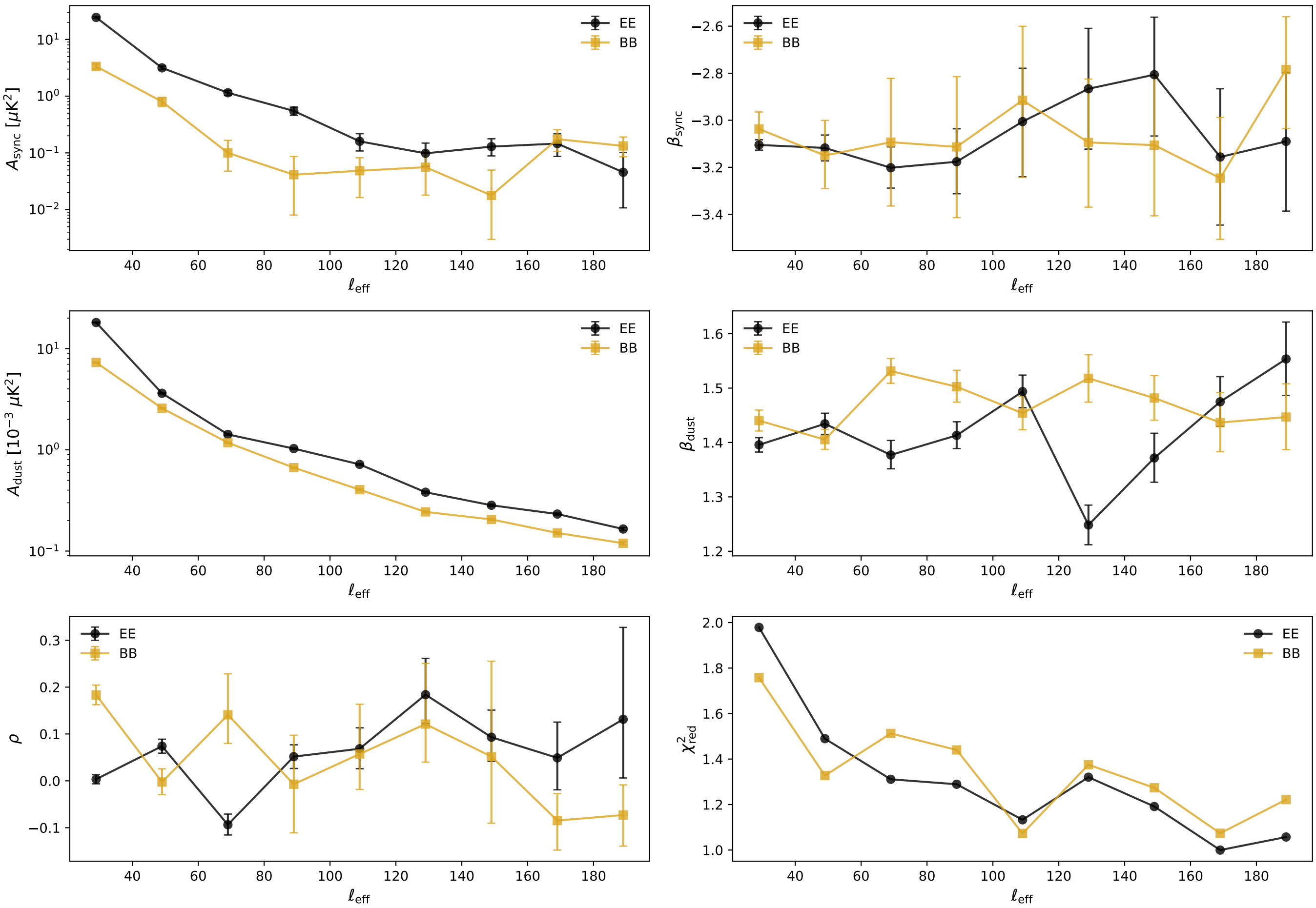

Figure 4: Bin-to-bin analysis results (observational) showing the evolution of fitted synchrotron and dust parameters as a function of multipole ℓ for both EE (top) and BB (bottom) modes.